



IMAGINE CUP



VEHICLE SPEED CONTROL USING GPS IN CRITICAL ZONES AND TRAFFIC DENSE AREAS

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In 2018, the UAE authorities reported **over 4,000** traffic accidents, resulting in hundreds of fatalities and thousands of injuries annually. (Source: Gulf News)

Patients suffering from osteoarthritis, rheumatoid arthritis, gout, or post-traumatic arthritis find it very **difficult to apply brakes frequently**.

A condition called **Jumper's Knee** is caused by repeated stress on the tendon due to lengthy drives. (Dr. Christopher Singh, 2011)

**A CRISIS IN
MOTION**

EXISTING LIMITATIONS

Cruise Control: Maintains a fixed speed.

Adaptive Equipment: Provides specialized mechanisms for disabled drivers.

Adaptive Cruise Control: Keeps a fixed distance from the car ahead.

Auto Pilot: Automatically steers, accelerates, and brakes within its lane.

Self-Driving Cars: Operates completely without a driver.



SYSTEM ARCHITECTURE AND WORKFLOW

1. The user enters their destination into our proprietary app, which then processes the route and identifies the key turns along the way.

2. An Intel RealSense Camera will be employed to measure the real-time distance between the vehicle and any surrounding cars.

3. A Raspberry Pi will collect data from both the sensors and use it to generate appropriate signals that activate the necessary actuators.

4. The car's ECU will utilize these signals to adjust the throttle (accelerator) and braking systems as required.





WELCOME TO CARDA

BlueTooth

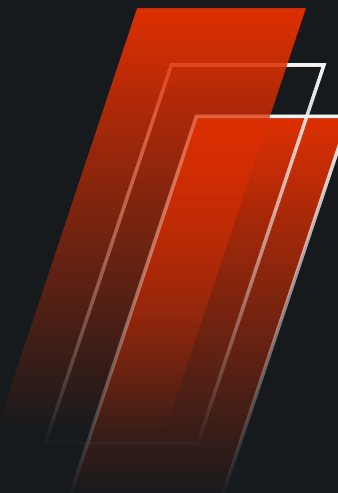
OPEN MAP

CAPABILITIES AND EXECUTION



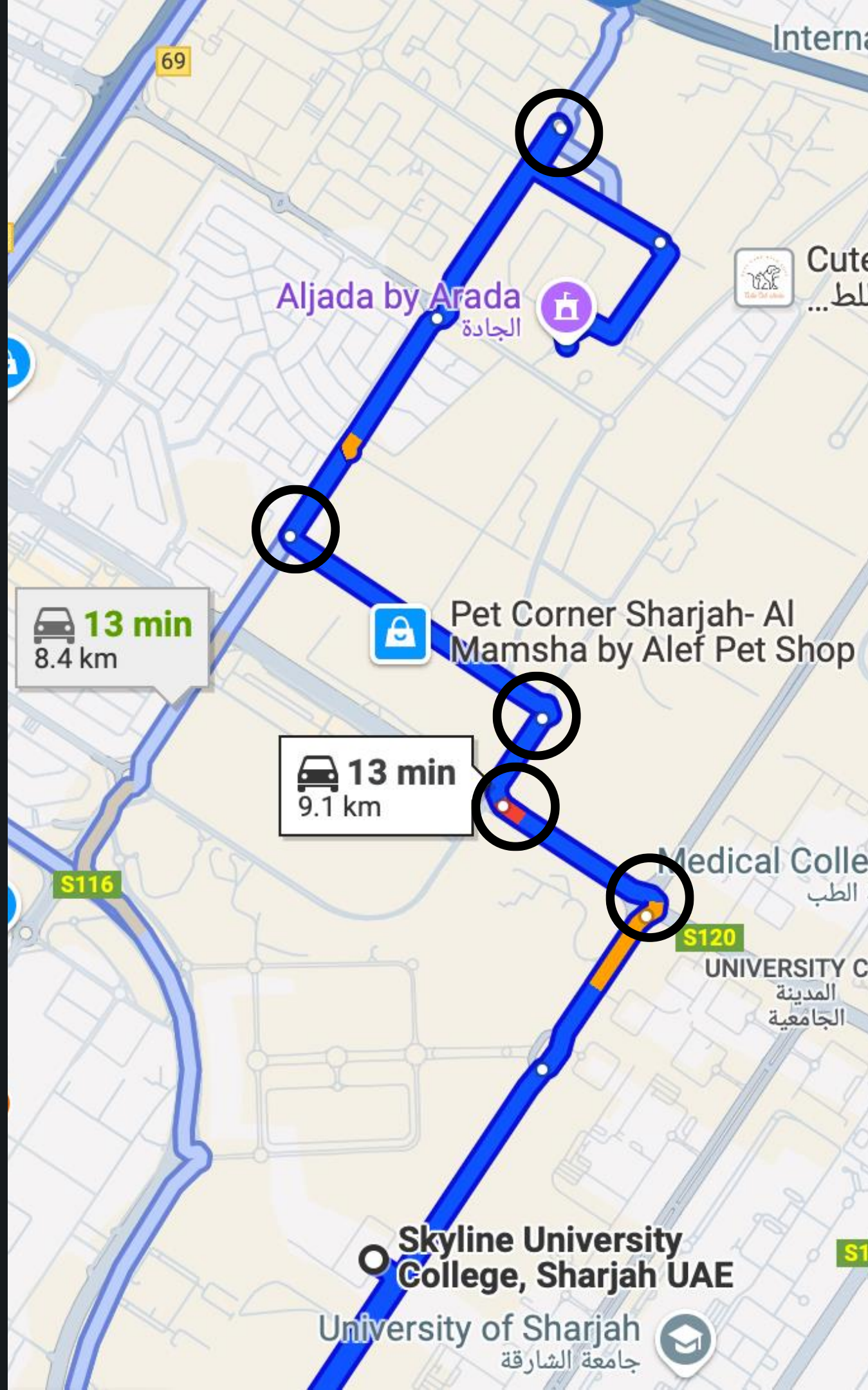
Detection of Critical Zones (Turns)

- Extract turn locations using **Google Maps API**
- Developed a mobile app in **Android Studio** for data handling
- Integrated **Google Maps SDK** for real-time turn detection and route visualization



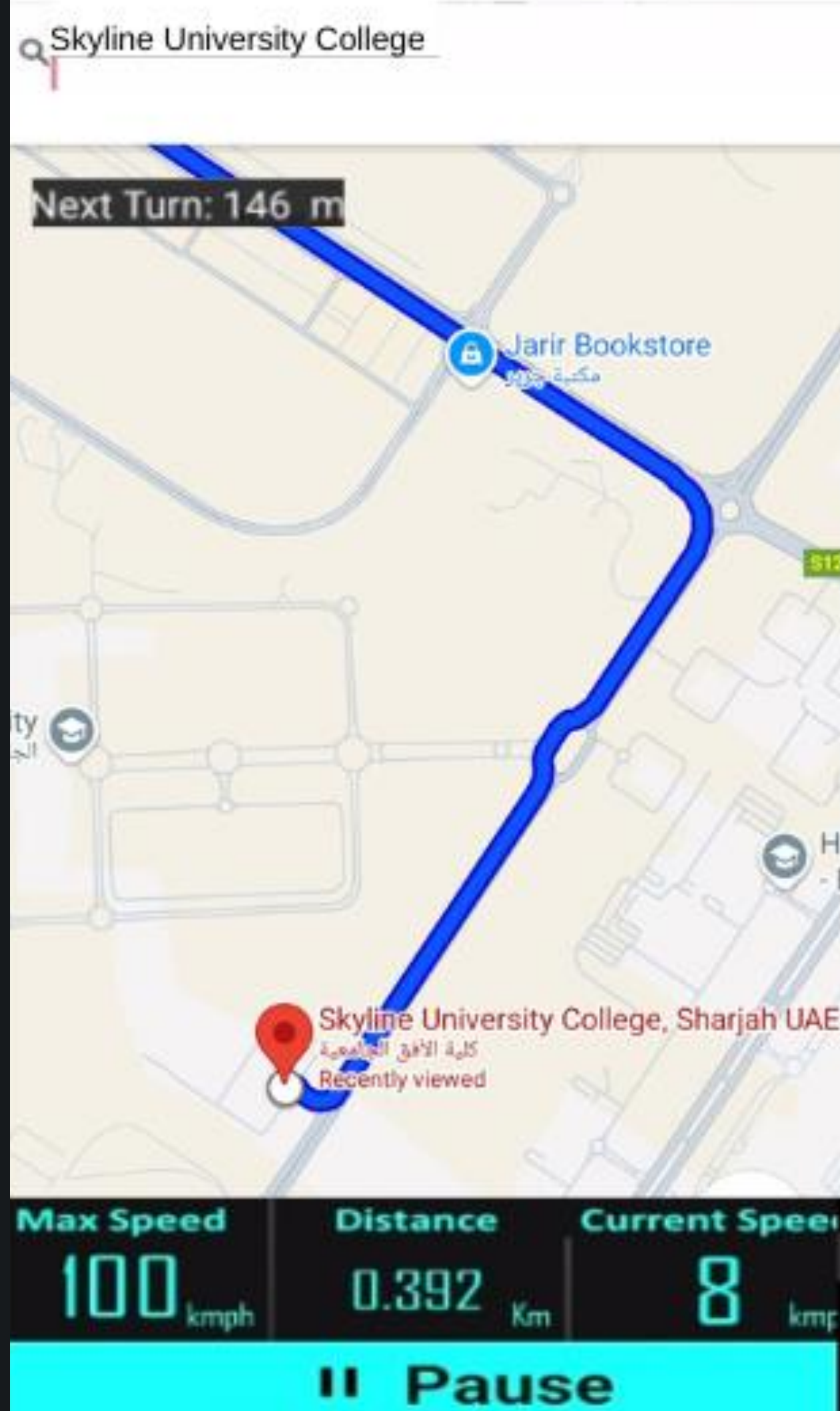
Google Maps API services

- Our app integrates maps through the Google Maps Class from the Google Maps SDK.
- The Google Places API assists users in locating their destination.
- The GeoCoding API transforms addresses into geographical coordinates.
 - The Directions API (DirectionsRoute) determines the optimal route to the destination.
- We use markers and polylines for clear visual representation on the map.



Getting Coordinates of Turns

- A waypoint consists of a single DirectionsLeg.
- The DirectionsLeg contains a steps[] array of DirectionsStep objects.
- A DirectionsStep holds all the details for one step of the route.
 - Each step has a Start_location and End_location in LatLng.
- Real-time distance is calculated between the current location and the step's End_location.
- This distance is sent to the Control Unit via Bluetooth.
 - A trigger is activated once the distance exceeds a set threshold.





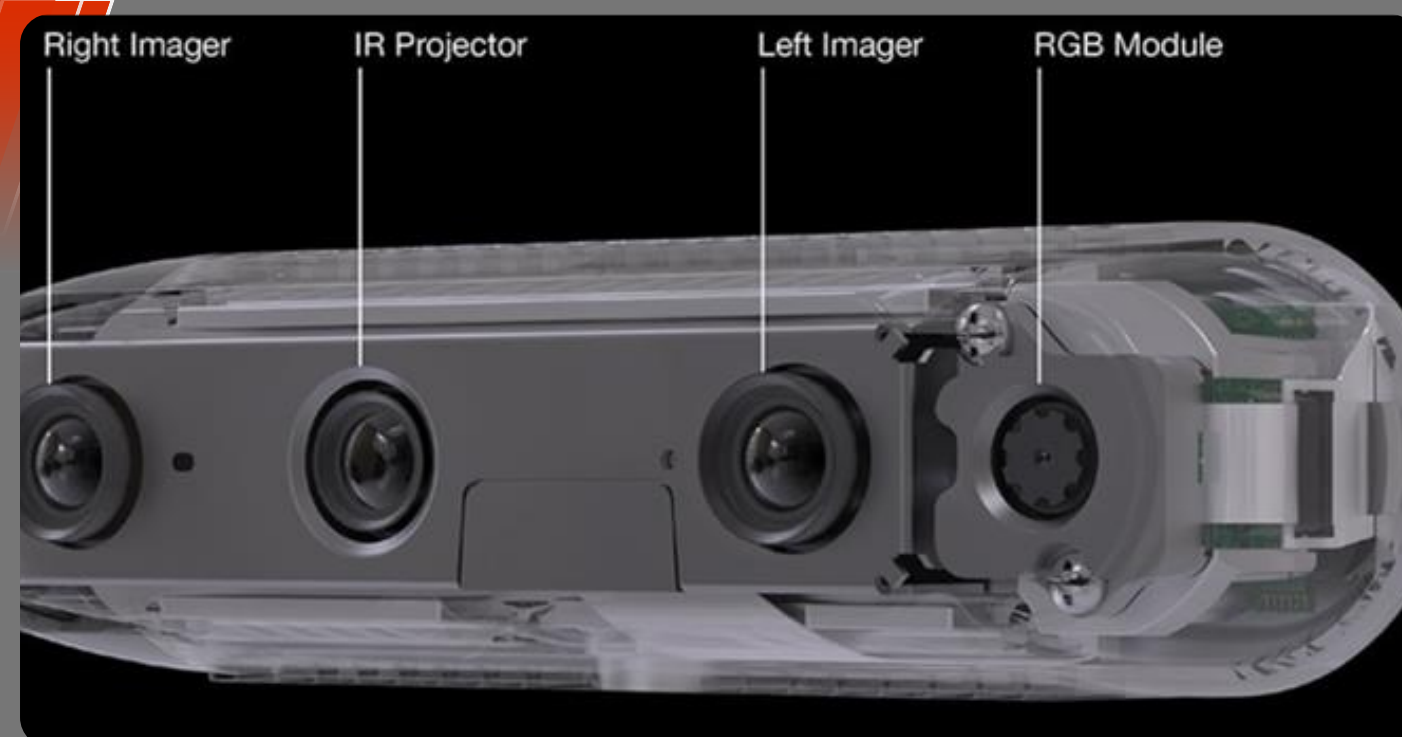
**VELOCONTROL
VISIONARIES**

Depth in Focus: Intel D435

RealSense Camera

The Intel RealSense camera leverages stereo vision technology to deliver precise distance measurements for objects in its field of view.

PRELIMINARY COST EVALUATION



Intel Real-Sense
Depth Camera D435

\$180



Raspberry Pi

Pi 3B+
\$35

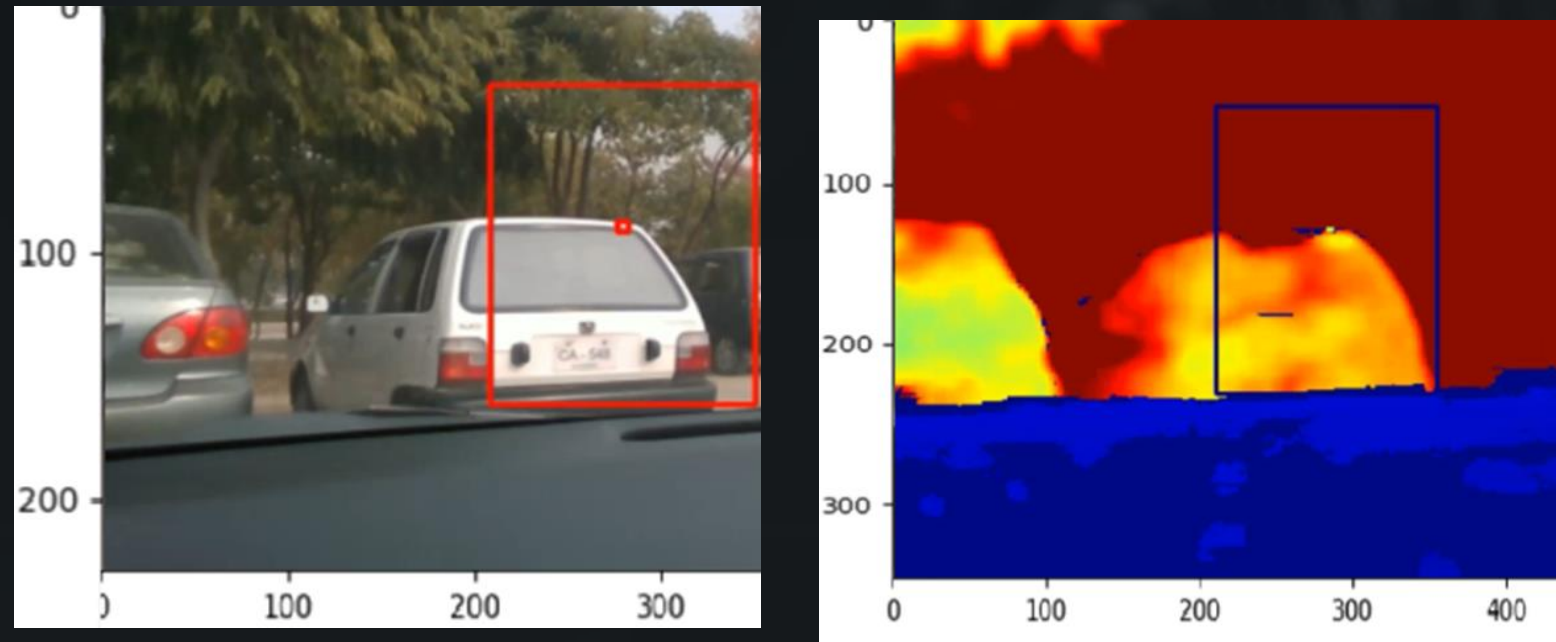


Actuators, Motors

\$30



MAPPING AND MEASURING DISTANCES

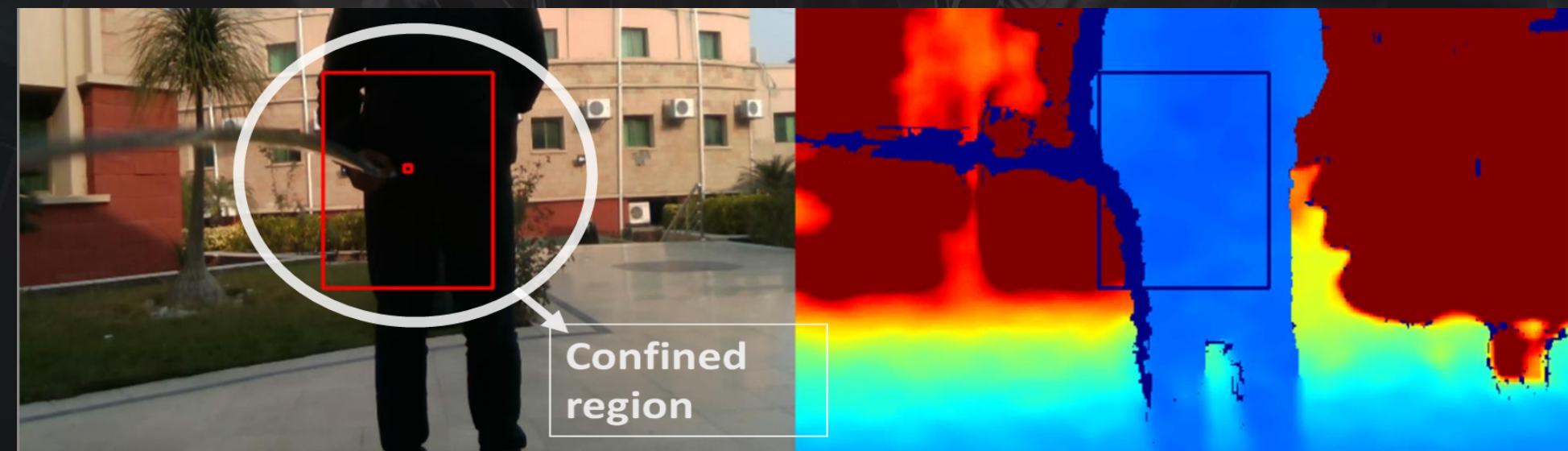


Camera through USB interface gives:

- Depth map (contain distance information of each pixel in it)
- Standard RGB image

ALIGNMENT AND CONFINEMENT OF FRAMES

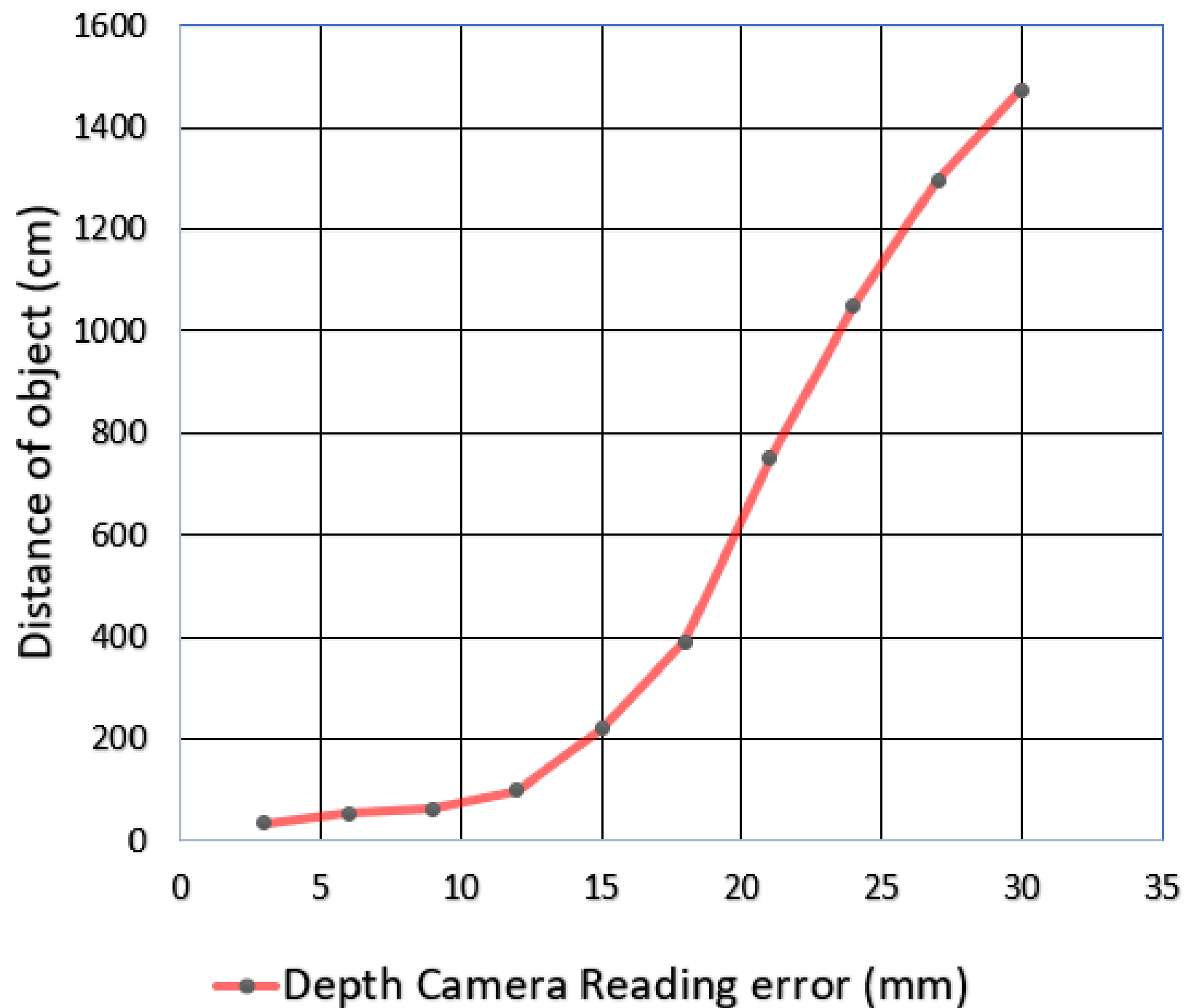
- To extract Distance we first have to align both Frames.
- To obtain distance of object in front camera, confinement on frame is done by drawing box over specific area.





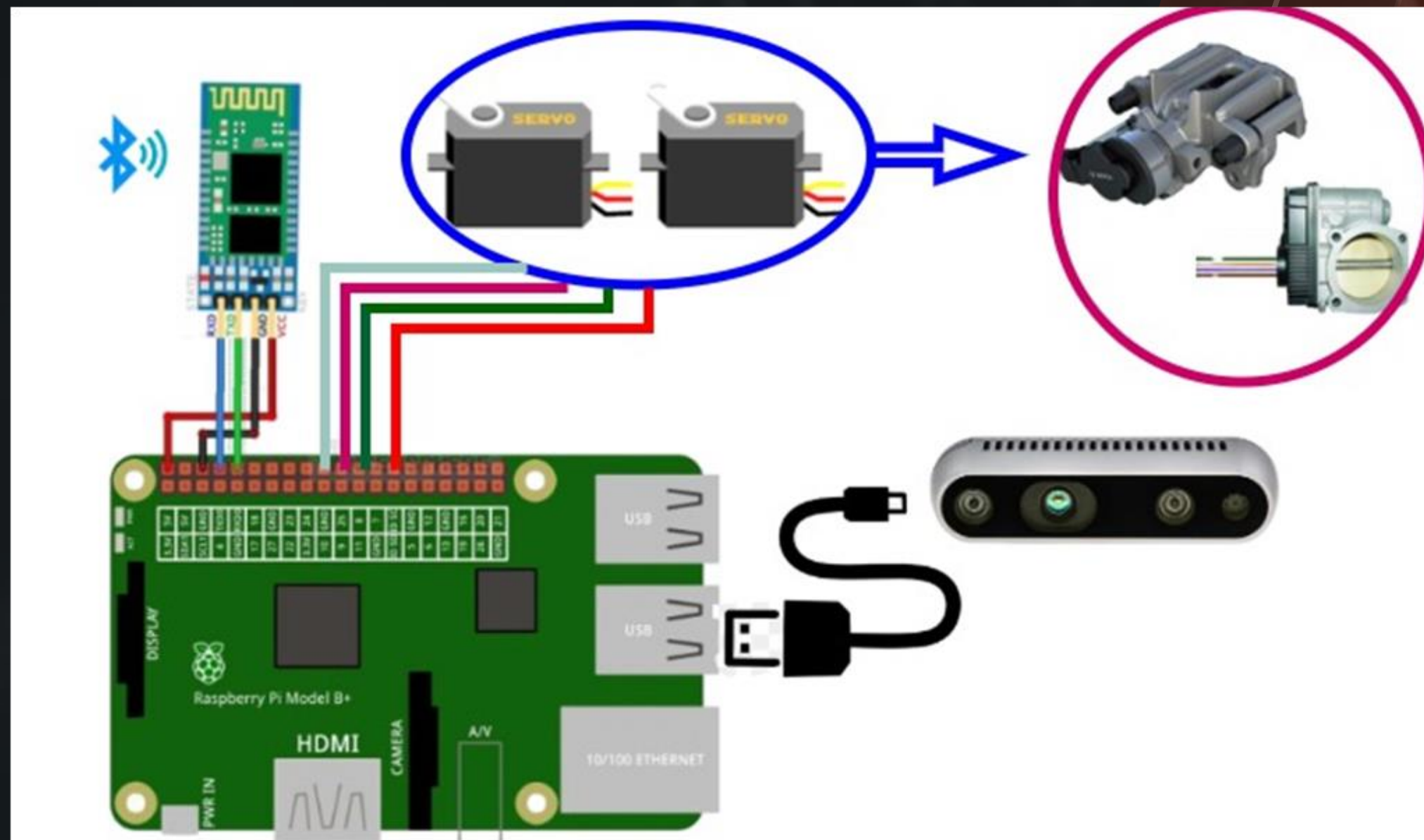
**VELOCONTROL
VISIONARIES**

RESULTS OBTAINED:



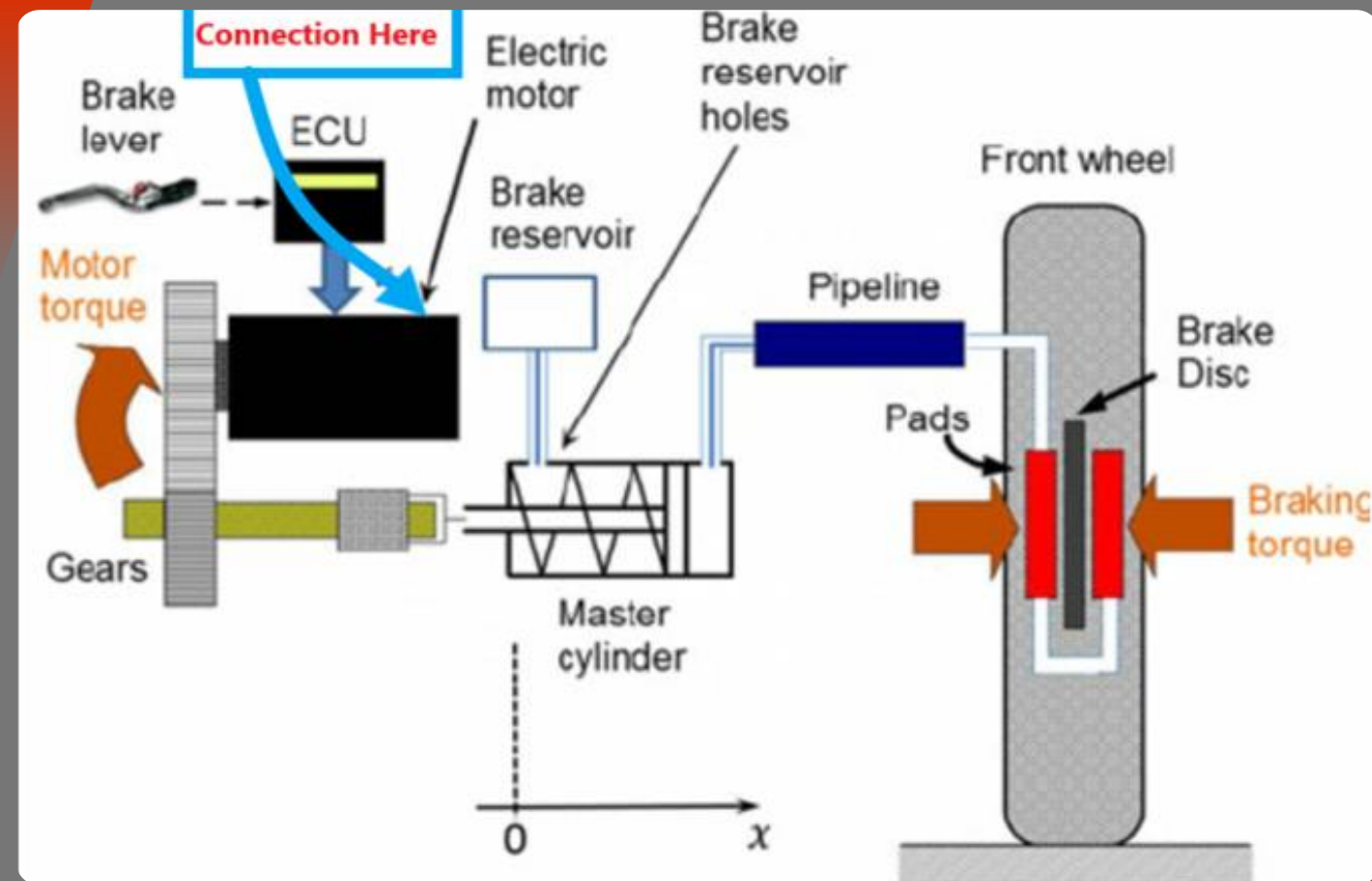
Distance (m)	Percentage Error
0-10	< 2 %
11-20	<3%
21-30	<= 5%

RASPBERRY PI WITH ADVANCED BLUETOOTH INTEGRATION



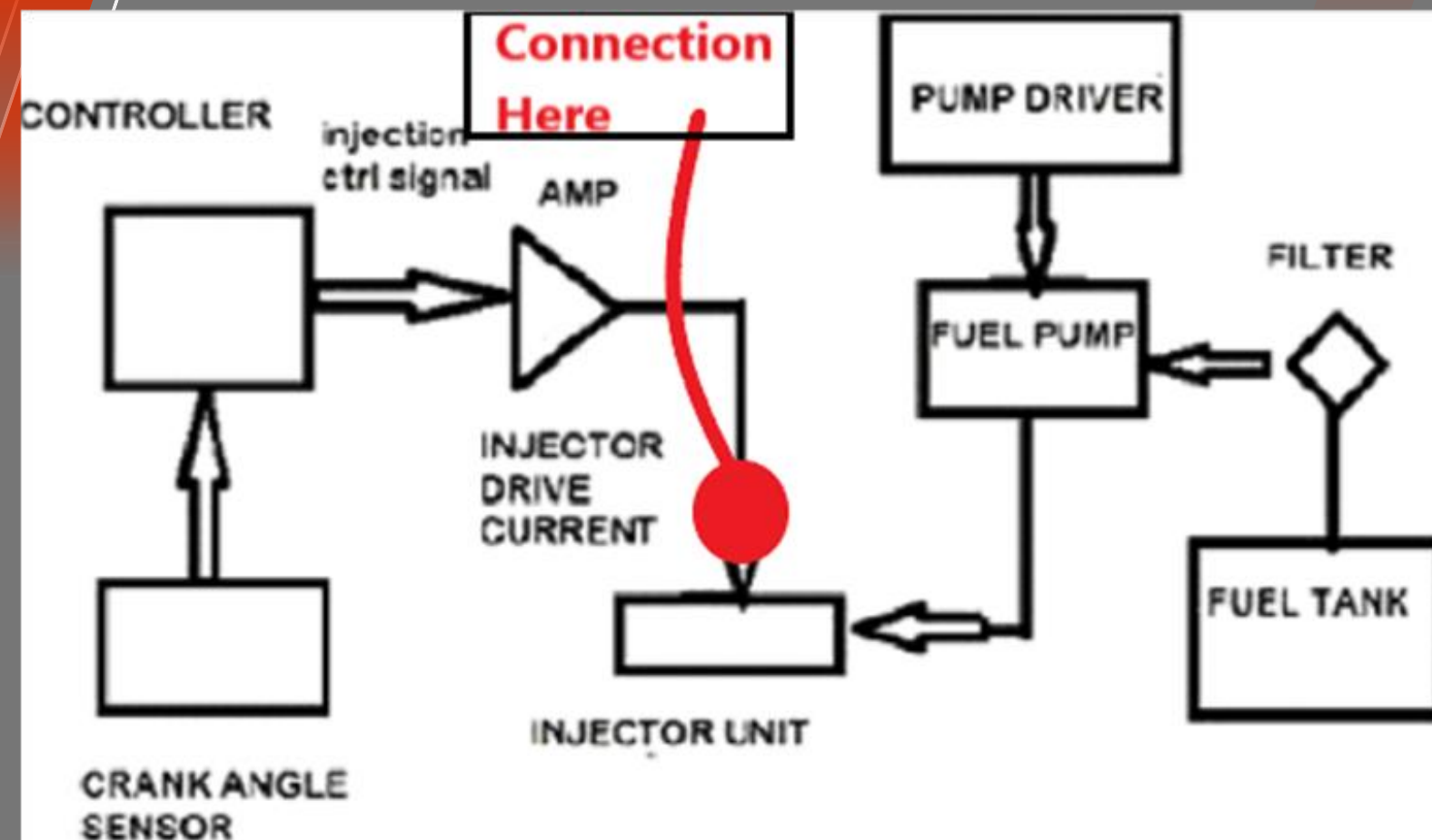
- The Raspberry Pi runs Raspbian OS with utilities and RealSense SDK 2.0.
- A USB interface connects the camera, while the Bluetooth module utilizes Serial Port 0.
- Processed distance data from the app and camera triggers PWM signals via GPIO pins.

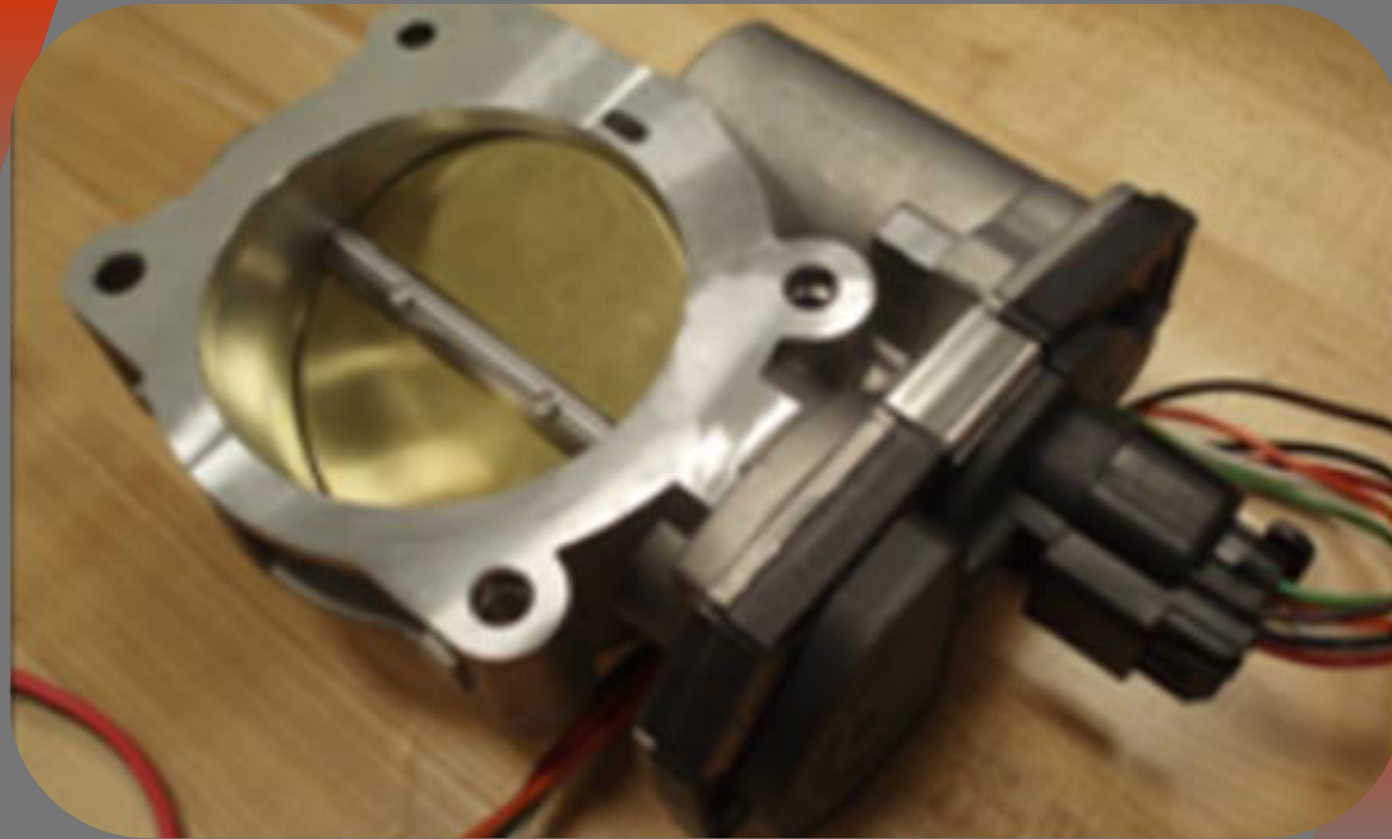




"System Integration with Automotive Electronics"

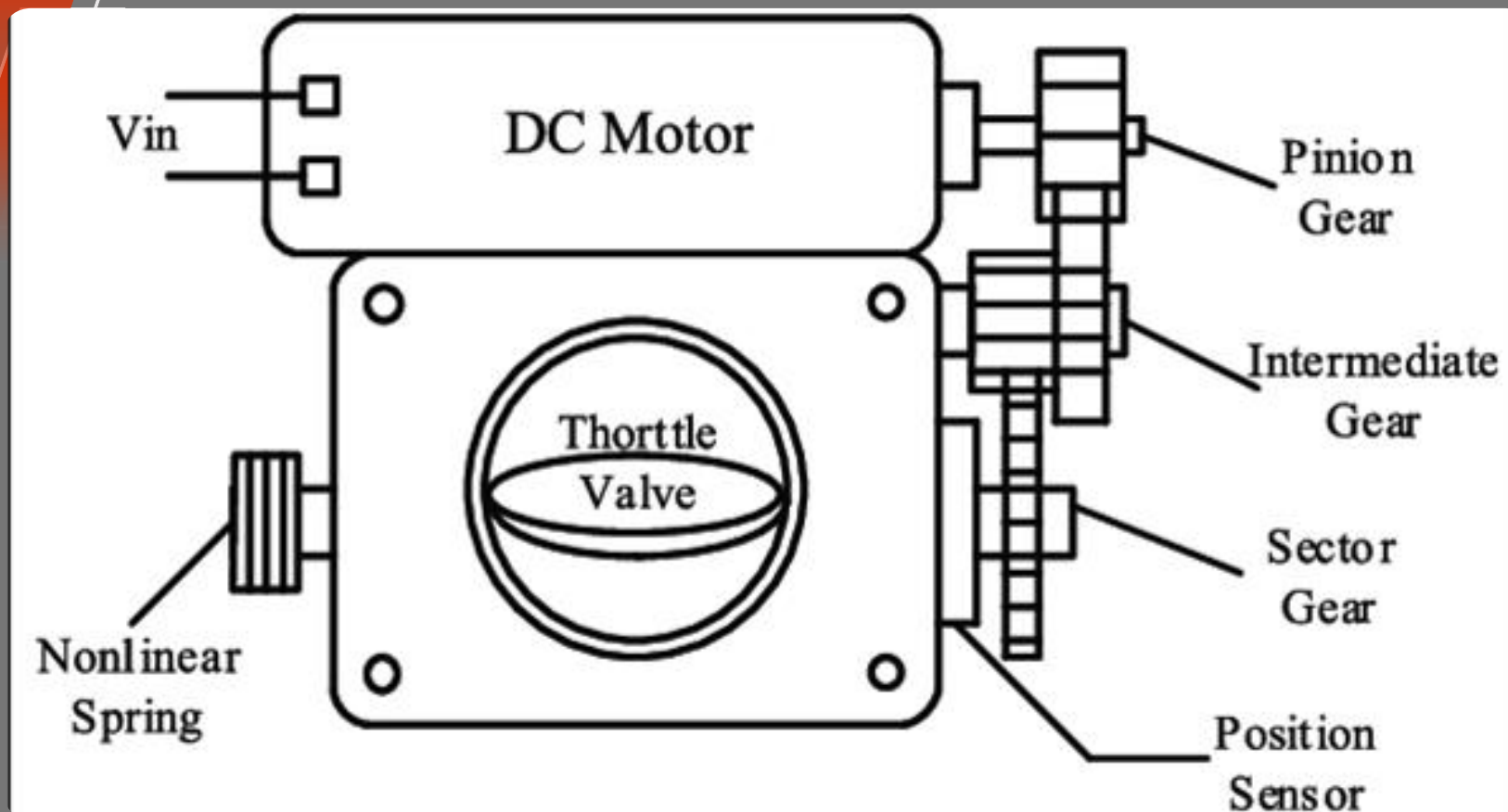
- The Electronic Control Unit (ECU) of the car manages all braking and acceleration operations.
- Both actuators for braking and acceleration include motors that operate via PWM signals generated by the ECU.
- The red markers indicate the connection points for our prototype in the block diagrams.





"EFI Redefined: Smart Fuel for Smarter Engines"

- In EFI systems, fuel delivery is managed by the Butterfly Valve within the Electronic Throttle Body.
- The Butterfly Valve is operated by a Servo Motor, as shown.
- PWM signals from our board effectively control acceleration and braking, enabling speed regulation.



CONCLUSION

Our system provides an innovative approach in the regulation of vehicle speed using GPS, advanced sensors, and intelligent algorithms that are seamlessly integrated. This improves road safety particularly in critical zones and high-traffic areas due to the precise proactive control of vehicle speed. Additionally, it supports efficient driving habits, and the likelihood of accidents decreases dramatically. The future development might be in 5G integration to provide better data transfer speed and AI-powered predictive analytics, allowing real-time adjustments for a better impact of the system on road safety.

THANK YOU!



REFERENCES

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